

Report

Quantum Mechanics in the study of Condensed Matter Physics

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Diptanuj Sarkar (22MS038)

DPS, IISER Kolkata

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1. Introduction

In this report, I will be attempting to synthesize a coherent narrative regarding the importance (or, lack thereof) of Quantum Mechanics or any other more “fundamental” discipline to the study of Condensed Matter Physics. To do so, I will be drawing from the ideas presented in the following literature:

- *Condensed Matter Physics: Does Quantum Mechanics Matter?*, by Michael E. Fisher [1].
- *There’s Plenty of Room in the Middle: The Unsung Revolution of the Renormalization Group*, by Nigel Goldenfeld [2]

After presenting the opinions from these authors, I will be presenting my (limited) skepticism regarding their claims. While Goldenfeld’s article does not go over much quantitative detail, Fisher’s article does as it is intended to be an overview of the state of Condensed Matter Physics when it was written. To keep the report from getting (unevenly) mathematically dense, I will attempt to frame the report focusing on Fisher’s general ideas rather than his specific examples.

Then, I will be attempting to present the current, decidedly quantum mechanical, questions that contemporary research in theoretical condensed matter is grappling with. This section will draw largely from [3].

2. Michael Fisher’s Ideas

In the article [1], Michael E. Fisher argues that Quantum Mechanics, as a discipline, is not really of “importance” to the study of Condensed Matter Physics. The article was meant to be an overview of the state of Condensed Matter Physics in 1988, and thus Fisher chose to utilize the overview to point out specific examples of how Quantum Mechanics is not of much relevance.

Note 2.1 (Fisher's Central Argument): Fisher states that each discipline of science has its own “language” and a corresponding “calculus” that is used to study the systems that the discipline is concerned with. Any such language or calculus can be shown to be derived from some underlying/“fundamental” discipline, but it is not *necessary* to do so as it does not provide any additional insight.

An easy to understand example that he supplies is that of the discipline of Sociology. It can be agreed upon by most people, via common sense, that sociology is indeed derived from the discipline of Biology - the biological necessities of organisms dictate their sociological interactions. But Biology itself can be shown to be derived from immensely complex systems that follow the laws of chemistry, or physics, or more progressively “fundamental” scientific disciplines. Despite this, it would certainly be absurd to state that Sociology *requires* the knowledge of Physics to be studied.

These, he claims, are effectively *different levels of discourse*. He relates this to Bohr’s idea of *complementarity* throughout the article, as it was meant to be written as a commemoration of Bohr’s 100th birth anniversary.

The idea of complementarity, as put forth by Bohr, is that certain properties of a quantum system are *mutually exclusive* but *equally necessary* to fully describe the system. What this means, in effect, is that the description of the system in terms of “classical” ideas such as particles and waves are revealed *due to* our choice of measurement apparatus - its “quantum” properties might be something else entirely. As a result, the

measurement apparatus and the system itself is essentially a single entity, and the “properties” we measure are the properties of this net entity, not of just the system itself.

The parallels between this and Fisher's idea are clear - it might be true that the problem being studied *can* be solved using the “fundamental” language of Quantum Mechanics, but it is not *necessary* to do so as it does not provide any additional insight. The language/calculus at a certain level of discourse regarding a particular system essentially provides an *effective* description that allows us to carry out calculations and derive quantities that are relevant to that particular level.

In Condensed Matter Physics, the main goal is to understand the varied manifestations of matter in its condensed states and to justify the existence and characters of the states using its “fundamental units”. In practice, this is most often done using *toy models* - caricatures of the actual physical systems being studied that manage to capture the essential physics. Such models, like the Ising model or the Heisenberg model seem to be guesses inspired via physical intuition rather than derivations from more fundamental principles. Even if they are derived, they almost all certainly gain a “life of their own” - the study of the properties of such models does not require the physical laws that might have been required for their derivations. Fisher goes ahead and claims that the wide ranging utility of such models is drawn precisely from their indifference towards the microscopic laws, as otherwise they would be *derived for a particular system* and lose their general applicability.

To Fisher, the main goal of a scientific discipline is to gain an *understanding* of the phenomena that it concerns itself with. Such an understanding is obviously best expressed in the language of that discipline - and such a language can not be imposed or decided upon without engaging in the discipline itself. In attempting to solve problems in any discipline, one might want to determine the relevant parameters and variables that they must study - but it is often the case that it is the *interactions* between these variables that captures the essential physics and not the identity of the variables themselves.

Since even the most “fundamental” theories are now understood to be low-temperature approximations of some underlying theory, this idea of Fisher's doesn't just apply to the “lesser” scientific disciplines, but to Physics itself. Since no ultimate “fundamental” science exists, and there is certainly no ultimate “derived” science, it can be argued that the study of how one can *go between* disciplines has more merit to it. Such a study of connections between different levels of description involve subtle ideas such as those Fisher himself worked on. As a result, he does express admiration and interest in this question. But despite this, he states that “... *they seldom add significantly to one's understanding of either the 'fundamental' starting theory or of the target discipline.*” There are cases where a model fails to capture *all* of the essential physics, and then a study of a more fundamental discipline helps shed light on what was missed - but this is an exception rather than the rule. To Fisher, the only meaningful reductionist project would be one that justifies the application of an intermediate model to a system using fundamental laws and not the *derivation* of all the physical properties of the system using fundamental laws.

2.1. The Stability of Matter

Fisher takes, for example, the relations between Statistical Mechanics and Thermodynamics. Thermodynamics is a discipline that deals with the description of the macroscopic properties of matter basing itself on a minimal set of basic *physical insights* - such as the existence of an extensive free energy function. Statistical Mechanics, on the other hand, is dedicated to the relation of microscopic mechanics to the macroscopic thermodynamics of matter.

Now, the “claim” of the existence of an extensive free energy function for bulk matter as stated in Thermodynamics *demands* that:

$$E_0(N) \geq -\varepsilon N \forall N \quad (1)$$

Where N quantifies some “size” of the system (number of particles, for example), $E_0(N)$ is the ground state energy of a system of size N , and ε is a finite constant that is independent of the size of the system. This is a statement of the *stability* of the system, and it essentially says that the free energy of the system must at least grow linearly with size.

While Bohr's theory of the atom states why an *atom* is stable, i.e. why the electron does not fall into the nucleus, it does not explain the stability of *bulk* matter. F.J. Dyson and A. Lenard were able to show that if we consider a bulk system of N particles that is neutral (i.e. equal number of opposite charges) and we allow for kinetic energy and Coulomb forces, the ground state energy has an upper bound:

$$E_0(N) < -\varepsilon_1 N^{7/5} \quad (2)$$

For some fixed $\varepsilon_1 > 0$. Since the exponent on RHS is greater than unity, it would appear that such a bulk system was not stable. But in a more detailed analysis, Dyson and Lenard show that this is not true *if* we account for Fermi-Dirac statistics.

They were able to show, that if the negatively charged particles were fermions and there were no more than q species of these fermions, the ground state energy satisfies:

$$E_0(N) > -\varepsilon q^{2/3} N \quad (3)$$

Which means that such matter is indeed stable. Note that this required the inclusion/consideration of Fermi-Dirac statistics, i.e. the Pauli Exclusion is just as crucial as the Coulomb force in determining the stability of bulk matter. It would appear that Fisher has contradicted himself here, but he goes on to state the opposite end of this stability problem - what happens to the free energy of a system of size N given that the particles interact via Coulomb forces and they are infinitely far away? We must require:

$$F(N) < \varepsilon_\infty N \quad (4)$$

Where $F(N)$ is the free energy of the system of size N , and ε_∞ is some constant that is independent of the size of the system. This would mean that Coulomb forces can not be *too repulsive* at long distances. The resolution to this problem lies in a completely *classical* fact - Coulomb charges screen, and thus the net interaction reduces. This requires only Newtonian ideas to show, and Quantum Mechanics is certainly not required to show this.

Thus, a claim that the very basis of the study of Condensed Matter - the existence of stable bulk matter - is dependent on the microscopic details of the system is simply false. At different limits, the justification of this stability requires one to appeal to ideas from either Quantum Mechanics or Classical Mechanics, but in doing so it simply demonstrates that neither is *essential* to justify the stability of bulk matter.

2.2. Polymers

Next, Fisher discusses polymeric matter - solutions of long polymer chains suspended in some liquid. Polymers have length, but they also have a finite thickness that must be considered for a meaningful analysis

of their properties. By virtue of this thickness, they can not be modelled as simple random walks. They must be modelled as a *self-avoiding random walk*. This translates itself into an “excluded volume problem”.

If we interpret the monomers that make up the polymer as the “steps” of a random walk, then the condition of “self-avoidance” means that the walker can not revisit a site it has previously visited. The mean size of a polymeric molecule is best expressed in terms of the *radius of gyration*, which is defined to be the square root of the average squared distance between all the monomers that compose the polymer. Because of the extended volume of the molecule, the radius of gyration grows as:

$$R_G \approx R_0 N^\nu \quad (5)$$

Where R_G is the radius of gyration, R_0 is some constant, N is the number of monomers/number of steps, and ν is a *scaling exponent*. It is known that for a random walk (non-self avoiding), $\nu = \frac{1}{2}$. Renormalization group ideas have led to the calculation of precise values of ν , depending on the dimensions the polymeric molecule inhabits. In 3-dimensions, we have $\nu \approx 0.588$.

Beyond the size, even macroscopic thermodynamic observables of such polymeric solutions obey certain scaling laws. Take for example the osmotic pressure π of such a solution as a function of concentration c . A “universal scaling theory” can be constructed for this and other such quantities of this system. It can be shown that:

$$\tilde{\pi} = \frac{\pi(c, n)}{ck_B T} \quad (6)$$

Which is the expression for the *reduced osmotic pressure*, with n being the number of polymeric molecules, follows a universal function of scaled concentration:

$$\tilde{\pi} \approx Z(x) \quad (7)$$

Where $x = \frac{c}{c^*}$ is a dimensionless variable, and $c^* = \frac{c_0}{N^{d\nu}}$. d is the dimension, and the scaling exponent ν shows up again.

This revelation leads to interesting predictions, such as the fact that the osmotic pressure (and other observables) of the solution become independent of the length N of the chain at high enough concentrations.

This theory has very wide applicability, over several decades of data. But never do we *require* a more fundamental theory to understand or arrive at this. Quantum Mechanics, which surely matters immensely on the scale of the monomers and even the polymer, do not show up at all in the derivation or motivations of this theory that applies so well to polymeric solutions.

As even Goldenfeld notes in his article [2], Fisher most probably went for the example of polymeric matter as it is prominent example of scaling laws showing up without any critical point. In doing so, the example serves to show that even in cases of matter away from the non-analytic behaviour at critical points, scaling laws and associated techniques can help glean information about the system without resorting to more fundamental disciplines.

2.3. Dimensionality over dynamics in magnetic matter

Moving on to the topic of magnetic states of matter, Fisher brings up ferromagnetism - the simplest case of a magnetic state. The existence of ferromagnetism depends crucially on the fact that electrons have spins, which surely makes quantum mechanics quite relevant as spins are quantal objects.

Historically, the first toy model that dealt with the origins of ferromagnetism was the Ising model - a simple model where the “spins” were allowed to have two discrete values and were coupled to their nearest neighbours in a way that encouraged alignment. This is the simplest possible way to model a system of “things” that “point” in a particular direction. Ising himself was able to derive an exact solution for the 1-dimensional Ising model, which showed that there was no ferromagnetic transition possible at non-zero temperatures.

A valid critique of this “simplistic” model would be that it fails to capture all the relevant details of electronic spins. Perhaps, in being so simple, it is providing us with an inaccurate prediction for 1-dimensional systems. A more sophisticated model would be the Heisenberg model, where instead of allowing for a Z_2 symmetry of up-down spins, spin vectors $\mathbf{S}$ were used with continuous rotational symmetry. The components of this vector followed the appropriate Pauli commutation relations, and they were coupled to nearest neighbours just as Ising did.

Despite this sophistication, the Heisenberg model predicts exactly the same thing as the Ising model did for 1-dimensional systems - there is no ferromagnetic transition at non-zero temperatures. In fact, the predictions offered by the more detailed Heisenberg model fares worse than the Ising model in 2-dimensions. Lars Onsager's solution for the 2D Ising model shows that there is spontaneous ferromagnetic order in the system at non-zero temperatures - this aligns with experimental observations in thin films of magnetic material. On the other hand, the 2-dimensional Heisenberg model *does not* have spontaneous ferromagnetic order at any non-zero temperatures.

The reason for this is as follows: most real-world magnetic materials do not possess the full symmetry represented in the Heisenberg model. The symmetry is broken via many microscopic interactions in the material, leaving the up-down symmetry as the only *effective* degree of freedom. So quantum mechanics was not only *unnecessary* for many of these systems, but the additional considerations it brought with it made the problem *harder* and less likely to represent the actual physics. The *important* features lie in the in “... *the interplay of entropy, fluctuations, and symmetry*”, as Fisher puts it.

Fisher emphasises this importance of dimensionality by replying on Peierls' non-rigorous argument for why the difference between the Ising and Heisenberg models show up in 2D. Let us deal with the Ising model first. If the model is to be stable in the ferromagnetic phase at non-zero temperatures, it must not have a reduction in its free energy if we include an “inclusion bubble” of spins pointing in the opposite direction. Let this bubble have a linear dimension of L , and so it will have a “volume” given by $\sim L^d$ and a “surface area” given by $\sim L^{d-1}$. The Free energy change $\Delta F = \Delta E - T\Delta S$ has the following contributions on the RHS:

- Bulk magnetic energy of the bubble: $\sim HL^d$
- Surface interface energy: $\sim +JL^{d-1}$
- Surface entropy: $\sim -TL^{d-1}$
- Positional entropy of the bubble center: $\sim -T \ln(L^d)$

For ordering to win out, ΔF must be positive for all values of L no matter how large. We require:

- Low enough temperatures: Roughly, $T < \frac{J}{k_B}$

- We must have $L^{d-1} > \ln(L^d) \Rightarrow d > 1$. This is the crucial dimensional condition that Fisher was referring to.

This holds for the Ising model because of how the boundary of the inclusion bubble looks - it is a sharp boundary. For the Heisenberg model, on the other hand, the boundary is a continuous region where the spins slowly rotate to smoothly join the two regions. This reduces the surface terms by a factor of L , leading to a $L^{d-2} > \ln(L^d) \Rightarrow d > 2$ condition for spontaneous ordering to be observed. For this argument to be shown rigorously, inclusion bubble should be allowed inside bigger bubbles, and so on - but the qualitative prediction remains the same. The Heisenberg model has *too much* symmetry.

2.4. The cost of $\hbar$

Fisher claims that the Planck Constant, something that is undisputedly related to Quantum Mechanics, can be shown to be an “irrelevant” parameter for systems near criticality at non-zero temperatures.

Let us look at magnetic systems again - the order parameter (in the Landau sense) for these systems can be understood to be completely classical “magnetisation” vector that has full rotational symmetry. If one wishes to relate this to the actual quantum mechanical reality, it could be understood to be the $S \rightarrow \infty$ limit where the spin values are so large that the quantized steps demanded by quantum mechanics are “washed out”. In the Landau paradigm, such magnetic systems can be dealt with entirely using Classical ideas.

To contrast, let us take the example of superfluids. Fisher admits that to locate the appropriate (Landau) order parameter for this system we do require certain insight from Quantum Mechanics - we require a macroscopic wavefunction. Such a wavefunction order parameter has the obvious magnitude and phase that we would expect from any microscopic quantum mechanical wavefunction. The $O(2)$ symmetry due to the phase can be mapped exactly onto a continuous rotation symmetry in the XY plane - the symmetry that the magnetic systems mentioned previously also possess.

Noting these similarities, it can be observed from calculations performed under the Renormalization Group framework informed only by these symmetries that the two systems are in fact in the very same universality class. Superfluid density of helium $\rho_s(T)$ vanishes according to the scaling law:

$$\rho_s(T) \approx D|t|^\zeta \quad (8)$$

Where $t = \frac{T_c - T}{T_c}$, and T_c is the characteristic critical point. Real interacting Bose gases, as well as XY magnets, have the exact same value of $\zeta = \frac{2}{3}$. $\hbar$ does not make an appearance, and it does not matter whether the system is classical (XY magnets) or quantal (superfluid) - the calculations give us the same result.

Fisher takes this idea and extends it, arguing that “*Planck's constant is worth only one extra dimension*”. Suppose the Hamiltonian operator for a system in d dimensions can be written as:

$$H = H_0 + H_1 \quad (9)$$

Where the eigenvalues $U(\sigma)$ (for a set of good quantum numbers σ) are known for H_0 . Since the quantum numbers themselves are classical variables, the evaluation of the partition function for H_0 is problem in classical statistical physics. The full partition function:

$$Z(T) = \text{Trace}(e^{-\beta(H_0 + H_1)}) \quad (10)$$

however, is difficult to calculate as $[H_0, H_1] \neq 0$ in general. This is precisely where $\hbar$ can find a way into the calculation. The way forward is given by Trotter's formula for the exponential of two non-commuting matrices:

$$e^{\hat{A}+\hat{B}} \approx \underbrace{e^{\hat{A}/n} e^{\hat{B}/n} e^{\hat{A}/n} \dots e^{\hat{B}/n}}_{2n \text{ factors}} \quad (11)$$

as $n \rightarrow \infty$. Putting this into the expression for the partition function for the full problem, it would appear as if we now have another classical statistical mechanics problem but with an increment in dimensions: $d \rightarrow d + 1$. What this means, is that instead of summing over a set of quantum labels σ , we are now required to sum over two labels to accomodate for the matrix elements of H_1 linking two different states. So, essentially, to deal with quantum mechanical systems *generally*, one has to only deal with a classical problem in one more dimension - quantum mechanics poses no new challenge in this context.

2.5. Modulated Matter

Fisher moves on from this to a discussion of some unexpected phases seen in materials such as Ag_3Mg - spatially modulated crystalline structure. The material usually forms a face centered cubic lattice with the silver ions on the cube faces and the magnesium ions on the corner sites of the unit cell. Now, if we take alloys of this material with a very similar composition, expressed as $Ag_{3-x}Mg_{1-x}$ with x being very small, something unexpected happens. Alternate planes of Magnesium atoms normal to one crystal axis are displaced by half of the unit cell diagonal with respect to each other. Two planes are shifted "up" this way, followed by two planes which are shifted "down". This justifies the naming of the phase - $< 2 >$.

This phase is clearly seen in micrographs of the crystal, and it can get even more complicated. These states are all characterized by spatial modulation of the structure with a periodicity that is larger than the ones imposed by atomic/ionic sizes. The obvious question here is how these large periodicities arise in the material.

Fisher admits the possibility of the properties of the ground-state quantum mechanical effects being relevant, but he dismisses it by saying that it is not adequate - the *true low-temperature* ground state might just be an ordinary fcc lattice. Since these long-wavelength spatially modulated phases are found at higher temperatures, Fisher argues that their formation depends on an interplay of (classical) entropy and fluctuations instead. Additionally, such spatial modulation has also been shown to exist in some magnetic substances, where the modulation shown up in the spin degree of freedom. The overall similarities between these magnetic materials and the alloys showing the same modulation seems to suggest that there are some common principles that are associated with this phenomena.

Having concluded this, Fisher retreats to the Ising model in 3-dimensions as the appropriate toy model, with a simple modification - the nearest neighbour coupling is retained but the couplings are assigned a value of J_0 in the x and z directions, and J_1 in the y direction. On top of this, an extra coupling of a spin with its *second neighbours* is put in, with a strength of J_2 in a particular direction. This model is dubbed the "Axial next-nearest-neighbour Ising Model" or ANNNI. Assigning $J_2 < 0$ makes the NNN coupling antiferromagnetic, which makes it compete with the ferromagnetic NN couplings.

Writing $J_2 = -\kappa J_1$, κ captures the "degree of competition" between the NN and NNN couplings. For $\kappa < \frac{1}{2}$ the ground state remains totally antiferromagnetic, but for $\kappa > \frac{1}{2}$ the $< 2 >$ -phase configuration becomes the ground state. No other phases can exist at zero temperature. But on raising the temperature, the interplay of entropy and fluctuations can easily be used to show that there is an infinite number of different spatially

modulated phases in this model. The concepts of effective interactions between domain walls is enough to explain this phenomena.

2.6. Quantum Hall effect

At the end of his article, Fisher turns to a description of the Quantum Hall Effect (Integer, and Fractional). Fisher here admits that “... *quantum mechanics comes back with a vengeance!*”, as it is clear that a Quantum Mechanical explanation in the form of Robert Laughlin's extended Landau Level wavefunctions, are required to understand the phenomena.

Fisher includes an description of the phenomena, which I will omit here for brevity. But in an apparent attempt to refer back to his overarching thesis of Quantum Mechanics being largely irrelevant for the study of Condensed Matter physics, Fisher uses a single-electron approximation to derive the quantisation condition of Hall voltage - something that he himself admits is not very accurate as it ignores electron-electron interactions.

In fact, at higher fields the Fractional Quantum Hall effect requires the inclusion of electron interactions to explain it, as the effect is fundamentally based upon the topological characteristics of the many particle wavefunction. The FQHE presents us with a fundamentally new state of matter, an *incompressible quantum fluid* whose fundamental excitations carry *fractional charges*.

Fisher here attempts to map the Laughlin wavefunction to classical Boltzmann weights by squaring the wavefunction - it can be shown that the Hamiltonian in the Boltzmann weights derived in this way correspond to a one-component plasma. This way, Fisher argues that even in the extreme case of FQHE the relevance of classical statistical mechanics is not gone - the ideas of entropy and fluctuation still have a “hidden role” to play.

3. Nigel Goldenfeld's Ideas

Nigel Goldenfeld's article [2], atleast in the first few sections, are directed as a retrospective commentary on Fisher's article [1], written nearly 35 years after the latter.

Goldenfeld begins by acknowledging Fisher's role as one of the first to adopt an “anti-reductionist” approach to the study of Condensed Matter. He admits to the significant shifts that have occurred in physics since the publication of [1] - but these shifts, in his opinion, have only validated Fisher's opinions. As a consequence, Goldenfeld's article is focused on the “middle scale” or level of description, not the microscopic. He defines this as the scale that is between the lattice spacings and the correlation length of a material - something which can be adjusted arbitrarily by tuning the temperature of the material near a critical point. It is at this scale, Goldenfeld states, where essential *cooperative* and *universal* phenomena lie. These phenomena may help us understand the “middle” space between disciplines of science, a region where “... *the greatest scientific excitement can often be generated*”.

To emphasize upon Fisher's opinions, Goldenfeld chooses two particular theories to discuss, as they are most relevant to Fisher's conception of what Condensed Matter Physics is supposed to have as its “task”:

- Bogoliubov's theory of superfluidity for weakly interacting Bose gas (1947)
- Bardeen-Cooper-Schrieffer (BCS) Theory of Superconductivity (1957)

A discussion of these theories and the states of matter they describe are related to the notions of “minimal models” and “levels of description”.

3.1. Minimal Models

According to Goldenfeld, both Bogoliubov's theory of superfluidity and BCS theory demonstrate adequately the surprising explanatory power of simple models of interacting bosons or electrons that choose to ignore many of the complexities of the actual physical system.

Bogoliubov, for example, assumed that the “weak interactions” between bosons in the superfluid were adequately described if we accounted for the “impenetrability” of the particles themselves, due to a sharp increase in interaction potential close to the particles themselves. This effectively amounts to replacing the real interaction potential (which could be very complex), with a contact interaction given by $U\delta(\mathbf{r})$. Bogoliubov understood that this approximation only held at long wavelengths, with respect to the atomic or molecular size scales.

BCS, on the other hand, took the Bardeen-Pines Hamiltonian that described electron-phonon interactions and added Coulomb effects to it. The approximation was made by assuming that “each state is connected to n other states by the same matrix element $-V$ ” - this technically applies strictly only to a subset of states that were paired with equal and opposite spin and momentum. This (approximate) theory was enough to capture the formation of the condensed state, for arbitrarily weak matrix element V .

Both the Bogoliubov and the BCS models are called “minimal” - models that are effectively only caricatures of the real world physical system, but which can capture the essential phenomena. Goldenfeld states that this idea of “essential phenomena” is a rather subtle one, and can be understood better through a discussion on Asymptotics

3.2. Asymptotics and Universality

Goldenfeld states outright that Bogoliubov and BCS' success was because they have adopted *non-perturbative methods* to extract physical conclusions from their models. BCS utilized a variational method, while Bogoliubov invented a canonical transformation that could be applied right after a naive perturbation theory approach has been applied.

This emphasis on non-perturbative approaches is made because the essential results from both the theories are *non-analytic* - these results are beyond the scope of regular, finite order perturbation theory. Due to their non-perturbative derivation, the results are believed to hold asymptotically as leading order approximations in the weak perturbation limit of $V \rightarrow 0$.

Goldenfeld emphasizes here on the non-analytic nature of phase boundaries in matter - the necessity of the non-perturbative techniques in his opinion are not due to the perceived “exotic” nature of the phases of superfluidity and superconductivity, but rather due to the non-analytic nature of the relevant partition function near the transition point. He argues that the emergent states near phase boundaries thus can not be simple perturbations around the normal state. The minimal model, in simple terms, captures the essential mathematical structure of the phase it is attempting to describe - all additional details on top of the model merely act as embellishments and thus show up as trailing, sub-leading order corrections to the predictions derived from the minimal model. By the very non-analytical nature of the critical point, Goldenfeld argues, no mathematical expansion about the point can be convergent and thus our minimal model results are at

best asymptotic *in the minimal model interaction parameter*. He stresses that these non-analytic results even show up at the mean-field level of description - fluctuations merely add on a “level of non-analyticity”.

Through this line of argumentation, Goldenfeld asserts that the study of the mathematical descriptions of phase transitions of matter necessarily require asymptotic methods - excess detail only complicates the task of extracting the asymptotic structure. These excess details often arrive from a microscopic description of the phase of matter, thereby implying that a microscopic description is effectively irrelevant.

In this way, a minimal model represents an entire universality class of models, and the asymptotic results derived from the minimal model via non-perturbative methods will hold regardless of the excess details.

Goldenfeld states that this use of minimal models as explanatory devices is “... *what physicists actually do*”, while admitting that it is an idea/approach that has been challenged in philosophical literature (which will be referred to again in Section 4).

3.3. Levels of description

Bogoliubov's theory shows that the fundamental excitations of an interacting system can be (to first approximation) treated as a collection of particles/excitations that act as a single non-interacting quasiparticle. In the case of superconductors too, these quasiparticles can be identified as Cooper pairs. For most purposes, superfluids or superconductors have a *universal* description that effectively hides the details and the interacting constituents of the matter that is in the phase.

In the case of Bogoliubov's theory, the parameter U that appears in the contact potential is actually dependent on the s-wave scattering length, derived from the collision of two bosons. The theory's approximation is valid (as stated before) only when this scattering length is much shorter than the mean separation between the bosons. In this way, one could say that U is a parameter that related two different levels of description of the system - it is the only “relevant piece of information” from the microscopic regime.

Goldenfeld uses this identification to stress of Fisher's opinion that “toy models” are supposed to play a central role in Condensed Matter. What is important is the identification of the relevant parameters to describe the model, not the *derivation* of the model from a detailed microscopic picture. He provides an example - The Navier-Stokes equations maybe derived from Boltzmann's kinetic equation, or it may be derived directly from phenomenological motivation. The latter is brief, and the former is quite lengthy and involved. From this, one would assume that the more “rigorous” Boltzmann-to-Navier-Stokes derivation would be more general, but that is not true as it is only valid in the low density limit - it is otherwise known that the Navier-Stokes equations can provide accurate predictions even in the high density regime. Thus, such “derivations” from microscopic theories only serve to reduce the regime of applicability of the model.

To end the sections of his article relevant to this report, Goldenfeld brings up the example of polymeric matter as detailed by Fisher in his article and discussed in Section 2.2. Goldenfeld states that he believes Fisher brought this up as an example of scaling laws (and thus, universality/asymptotic behaviour) showing up in systems away from criticality. He states that it was in the study of the conformations of polymers that the use of renormalization group theory first occurred to Wilson. Field-theoretic methods were also developed by Edwards in this context, which ignored fluctuations and had the same long-wavelength (in this context, this would mean that the length of the polymer is much larger than the lengthscales of the monomers) regime as Bogoliubov's theory. Subsequent development of configuration-space renormalization group methods by Ohta and Oono in the early 1980s finally led to a description of polymeric matter that could not only calculate

simple scaling laws, but also calculate the universal scaling functions for the statistical thermodynamics of polymer solutions as a function of concentration.

4. My criticisms of Fisher and Goldenfeld's views

This section was re-written after my discussion regarding the articles [1], [2], [3], and [4] with Siddhartha sir, and includes a more critical response to the ideas put forward by Fisher and Goldenfeld. As a consequence, I will sometimes refer to [4] and my report on the same.

4.1. Summary of Fisher and Goldenfeld's arguments

To begin with, let me list out the main arguments that Fisher puts forward, as perceived by me. Since Goldenfeld's view only elaborate on Fisher's own (with the additional emphasis on levels of description and RG), I will subsume them into the same list:

1. Scientific disciplines operate on different strata of description - they each have their own "language" to describe their systems, and a "calculus" to derive useful results and deduce facts about the system.
2. In the context of Condensed Matter Physics, the "language" is provided by minimal toy models, and the corresponding "calculus" is Classical Statistical Mechanics - not Quantum Mechanics or any other "more fundamental" theory. The essential physical phenomena in various states of matter can be explained by studying toy models that capture the effective physics of the system, using statistical techniques.
3. A derivation, or justification, of a model from "first principles" (i.e. the microscopic details of a real physical system) is not required - it does not add to our understanding of the system in most cases and only serves to reduce the regime of applicability of the model. The goal is to "understand" and not to "reduce".
4. Macroscopic "ordering" is dictated by classical concepts like symmetry, entropy, and dimensionality of the system, not by Quantum concepts like entanglement. This can be thought of as an elaboration on Point 2.

The clear anti-reductionist stance taken by both authors is something that I can agree with - it is very clear that all of Physics, no matter the discipline, is only an *approximate description of nature*. In such a scenario, a purely reductionist approach is futile. There is no bottom of the barrel when it comes to Physics. Even if there were a level of description "fundamental enough" for all our needs, it would be a monumental task to derive and utilize it to make predictions about the world around us. In addition, of course, it requires just as much work (and novel ideas) to explain emergent phenomena as it requires to deduce fundamental laws. This is explained beautifully by P. W. Anderson in his article, "More is Different" [5].

In agreeing with the authors on this point, I would like to stress that there are clearly several caveats to such an approach. It is true that a lot of progress, technological and otherwise, has been achieved by effectively embellishing minimal models with the necessary amount of detail. However, it has become increasingly clear over the last few decades that a rigid unwillingness to borrow "language" and "calculus" from more microscopic disciplines can (and has) lead to profound organizational confusion in the study of Condensed Matter. Both Fisher and Goldenfeld seem to be quite optimistic about progress in Condensed Matter utilizing minimal models formulated out of classical ideas, which is something that they have not convinced me of.

4.2. The GLW Paradigm, and why it fails

Fisher was writing at a time when the Ginzburg-Landau-Wilson (GLW) paradigm was showing its limitations, but was still firmly placed as the basic tool that Condensed Matter Physicists used to describe and explain

states of matter. The symmetries of the system were determined, the local order parameter was selected, and a functional expansion was carried out in terms of the order parameter and its gradients in a way that the functional remained invariant under the symmetry transformations that the system respected. Then, using the idea of Spontaneously Broken Symmetry (SSB), we were able to *justify* the properties of the low-temperature state. Utilizing ideas of adiabatic continuity justified the study of minimal models. The Goldstone theorem, and the Anderson-Higgs mechanism supplied us with a definitive way to understand the low-energy excitations of ordered states by simply using the symmetries of the system. While Landau theory failed near the critical point, Wilson's RG theory could remedy the situation by describing the "flow" of the coupling constants in Ginzburg-Landau functionals in theory space - supplying a way to precisely calculate critical exponents and justify the existence of universality classes. The GLW paradigm is thus a successful, and most importantly a *cohesive* way to describe Condensed Matter systems (up until then).

In addition to this, progress in closely related disciplines like Many Body theory, QFT and Statistical Mechanics seemed to provide a passage between the intricacies of each individual model. In many cases, as Fisher demonstrates, well known models could be adapted via slight alterations to demonstrate the emergence of a variety of states (See ANNNI in Section 3.1.).

But, before 1988 it was increasingly becoming clear that GLW was not enough to describe the myriad states that experimentalists were encountering. The entire premise of the GLW paradigm rests upon the ability of the physicist to accurately deduce the effective symmetries and local order parameter of the system. Without such an identification, it is not possible to use the GLW paradigm as a framework to describe and explain the state.

One of the first examples of this failure is the Quantum Hall Effect, which Fisher mentions at the end of his article. Fisher tries to show a mapping between the Laughlin wavefunction and Boltzmann weights, no doubt trying to support his thesis of Quantum Mechanics being irrelevant in the discussion. But clearly, a mapping such as this neither has the explanatory power, nor the predictive power, that Laughlin's theory does - it is in fact more of a sidenote than a theory. Even if it were fleshed out to its logical conclusion, one could wonder if the explanation that Fisher provides could even be conceived of without first being aware of Laughlin's theoretical formulation. Without concepts of bulk-boundary correspondence, Anderson localization, and the language of Quantum Mechanics, it is not obvious how one should go about describing the state and constructing a theory for it. No symmetries, in the traditional sense, are broken in this state. There is no *local* feature of the state that can help us distinguish it from any other. An effective description of the QHE requires the language of topology to capture global features.

The High-temperature cuprate superconductor phase diagram is a diagram that encapsulates a lot of what is wrong with the "classical physics only" approach that Fisher advocates. Several states that have been identified in this material still resist a complete explanation, even after decades of study. More examples will be provided in Section 5. Several states of matter exist where the basic explanatory models that have been developed have to borrow heavily from the language of Quantum Mechanics and QFT.

A fair question here would be: Is it possible, at all, to create explanatory models for even these states using purely Classical ideas?

I remain unconvinced that such a thing is possible. Drawing from literature on the philosophy of science [6], [7], I find it convincing that explanatory and predictive powers of a model are drawn from certain *common features* that put the model and the real world systems it describes in the same "class". It is not a mapping of the

degrees of freedom of the model to that of the system that achieves this - it is solely due to shared fundamental features (like locality, or conservation laws). It could be possible to construct “classical” systems that exhibits the same macroscopic physical phenomena as a beyond-GLW state - but a prediction is not an explanation. To admit that a model that simply displays the same behaviour as a real world system somehow “explains” the latter is to admit to a sort of “explanatory asymmetry” [7], i.e. we are providing undue preference to the classical model’s explanatory power by virtue of it simply displaying the same features. Why not claim that the real system explain the model, then? Surely, the fascination with minimal models is driven by the epistemic/computational ease that they provide us in comparison to the real system(s). But despite the “ease” a model provides, we can not call it can “explanation” for the real world system without justifying *why* the microdifferences between the model and the system do not matter.

More than just “working”, an explanation requires a model whose construction gives us a sort of intuitive understanding of the physics. It must have degrees of freedom that retain some representational similarity with the real world systems that it seeks to caricature. It turns out that the most successful models to date draw heavily on the language of Quantum Mechanics to provide this intuition. A good example of this is Resonant Valence Band (RVB) models, where the concept of a singlet state being maximally entangled is absolutely crucial to the construction of the model itself. RVB models are ubiquitous in studying Quantum Spin Liquids. Kitaev’s chain and honeycomb models similarly require the idea of the Majorana fermion to be understood. Thus, it is my current opinion that demonstrating a mapping of a known quantum model to a classical one can not justify the optimism that Fisher and Goldenfeld display.

4.3. The lack of organizational principles

Beyond the question of relevant “language”, both authors fail to admit a glaring gap in the organization principles available to us in Condensed Matter Physics today. GLW was useful because it provided a definite roadmap to analyzing a state of matter. Now, all we possess to explain most states are inspired minimal models. An overarching organizational principle, like GLW, is missing in modern Condensed Matter Physics - we have no way of knowing how many quantifiers are required to be specified for a state for us to be able to understand it.

One would expect that a natural *extension* to the GLW paradigm can be invented, so that the newer paradigm would be able to classify, explain, and predict these new states of matter on the basis of their natural language - quantum mechanics, topology, gauge theories, etc. while still retaining the same explanatory power for the states that classical physics describes. Perhaps such an extension can be achieved via a generalization of the two main features of GLW - symmetry and local order parameters. The former could perhaps be generalized by including “higher” symmetries (belonging to a more general algebraic structure than the typical group structure) that could be understood as “broken” in some states, while the latter could be redefined to capture *global* characteristics of the material instead of just local ones. An inclusion of an additional essential quantifier may also be required to capture relevant information on the geometrical/topological features of the state. While this is just speculative, it can not be denied that there is a need for progress in this direction.

It appears to me that instead of pursuing a sort of conservative approach to the language of the discipline (i.e. pursuing an attachment with the classical language), there is a need to embrace new languages from whatever discipline it may be given that it allows us to describe these beyond-GLW systems in an intuitive fashion. Whether it be Quantum Mechanics, QFT, or mathematical disciplines like Topology, or Graph Theory, an effort must be made to track down relevant ideas that can help describe these systems.

4.4. Acknowledging the importance of non-perturbative methods

Additionally, I would draw attention to Goldenfeld's insistence on non-perturbative methods being crucial to understanding asymptotic behaviour. In agreeing with his views of the same, I express agreement with Goldenfeld's mathematical intuition and *not* with the extent of his anti-reductionist views, as mentioned previously. This is clearly important for understanding the beyond-GLW states of matter discussed - the strong interactions in these states lead to situations where the kinetic and potential energy scales become equal, making perturbative techniques fail. Non perturbative techniques are thus crucial for making headway into the theory of beyond-GLW states, as they can lead us to effective theories that facilitate actual calculations to match experimental observations. RG methods serve to justify why a particular minimal model is in the same class as a real physical system, which is very important in justifying the explanatory power of the model. Since more and more models defy perturbative methods, non-perturbative techniques are critical in working with them.

5. Quantum 2.0

Several of the ideas presented in the previous section are drawn from a closer reading of [3], and I will not be summarising them again. I will, in this section, seek to summarize the current challenges that Condensed Matter physicists have to face in their research and the overall goals of the new vision for the field - Quantum 2.0.

As mentioned in previous sections, the physics of several states of quantum matter can not be understood in terms of non-interacting quasiparticles. As a result, important theories like Landau's Fermi liquid theory fail in many of these states of matter. The reason for this failure most lies in the inability to apply adiabatic continuity to label quasiparticles with quantum numbers, and the inability to find broken symmetries from which the quasiparticles emerge.

Important examples here are the gapless non-Fermi liquids, and gapped liquid-like insulating states, both showing up in the cuprate phase diagram (among many other systems) as mentioned before. These states display signs of topological order, and thus lay beyond the GLW framework. Many quantum materials also show signs of quantum criticality at finite temperatures, which may be linked with the breakdown of ordered insulating states (which may be classical symmetry broken states, or topologically ordered). These poorly understood metals may also serve as parent states for new kinds of unconventional superconducting states - which raises their importance from the the point of view of technology.

Disorder is seen to be very important in several beyond-GLW states, such as the QHE, but very little is known about what role it plays in general. Similarly, very little is known about how out-of-equilibrium dynamics shapes the features of quantum matter. Entanglement encoded into the eigenstates of these quantum materials has also not been entirely connected to its properties. These features are clearly important quantifiers of a state, but they have not been incorporated into some overarching framework yet.

A lack of such an overarching framework also shows up in the form of several unanswered questions:

- The melting of Mott insulating many-particle states and the formation of a Fermi surface with proximate gapless excitations can not be explained using perturbative methods.

- It is not understood what causes Landau quasiparticles (the fundamental entities in Fermi liquid theory) to break down and be replaced by some other form of gapless fermionic excitations near the Fermi surface in non-Fermi liquids (in 2 or more dimensions).
- Only a few features of topological order are known, and without the framework it is difficult to determine whether additional features exist and/or are relevant. The known features are:
 - Degeneracy of the ground state, dependent on the topology of the spatial manifold upon which the system is placed.
 - Topological excitations that can carry fractional charges, and have anyonic statistics.
 - The presence of edge states (like the current carrying edge states in QHE).
 - The existence of long-range entanglement.
- We do not possess an accurate understanding of the competition between the tendencies of a system of quantum spins to order magnetically, and be screened by interactions with a reservoir of conduction electrons.

Most of this confusion is neatly represented in the typical cuprate phase diagram mentioned several times before, but even this in itself is a question. It appears that the qualitative features of this diagram shows up in several other systems, and it is not entirely understood why that is.

5.1. Requirements for further progress

Standard numerical methods available to physicists, like exact diagonalization, are not adequate for “large” systems owing to the exponential growth of the Hilbert space with system size. Even with modern computational power, such an exponential growth can not be tamed, other methods must be sought out. Other methods face other difficulties:

- Quantum Monte Carlo: Sign problem at lower temperatures restricts it to higher temperatures.
- Density Matrix Renormalization Group: Restricted to 1-dimensional systems.
- Dynamical Mean Field Theory: Lack physical insight due to the opaque self-consistency procedure.

Several other methods can help us reach only perturbative regimes, in various parameters. In line with the stress put upon non-perturbative methods before, there is a clear need for the development of new non-perturbative analytical and numerical methods that effectively deal with these problems. It is also important that these methods help us identify the effective field theories that can describe the states, alongside capturing the physics of the quantum fluctuations that drive quantum phase transitions in states. The development of numerical methods can take guidance from a deeper understanding of first principles calculations of electronic bandstructures, as well as quantum simulators like cold atom systems.

But most importantly, there is a need to develop some universal principles that are relevant to the arising of emergent phenomena in quantum matter. Once such principles are constructed, theorist will have considerable ease in bridging the gap between their intuition regarding the physics, and the complex materials that they wish to study - we need to know *what* to look for.

Advances in experimental out-of-equilibrium dynamics at ultrafast timescales has given us access to non-equilibrium systems. If broad theoretical advances are made in the field of non-equilibrium statistical mechanics, then it might shed light on the problems of quantum matter. The connection between many-particle entanglement and the experimentally observable properties of quantum matter are also not yet understood well, and progress on this front can help us classify the various emergent states of quantum matter and help improve technological uses of quantum matter.

All such theoretical progress must be evaluated by:

- How much physical insight they provide.
- How accurately they explain the phenomena at hand.

Questions of mathematical elegance are simply irrelevant, and can be treated as a bonus (if it is there).

5.2. The situation with cuprates

An important milestone for progress would be the discovery of some explanatory model for the cuprates, that allows us to derive effective theories for the states of matter observed in these materials. Two models show promise:

- The 2-dimensional Hubbard model: Inspired by the effectively 2-dimensional physics of the cuprates, a Hubbard model with three simple parameters for on-site repulsion, hopping, and chemical potential in 2 spatial dimensions shows promise.
- The t-J model: This model comprises of nearest neighbour spin exchange interactions between *localized* spin-1/2s, and correlated hole hopping with a condition for forbidden double occupancy, which may be derived from a Hubbard model in the limit of very strong on-site repulsion.

Despite focused effort, these models have only helped us understand very little of the physics of the cuprates - it has not been able to extract effective theories from them. This suggests that the bottleneck is a lack of non-perturbative methods required to study these models.

6. Conclusion

I will end the report by listing out, again, a few broad questions that can guide future theoretical progress as stated in [3]:

1. Is it possible for us to build a unifying theoretical framework for understanding how electronic correlations manifest in the interplay of the charge, spin, and orbital degrees of freedom, that lead to various ordered states of quantum matter?
2. Is there a connection between non-Fermi liquids and unconventional superconductors? If yes, will an understanding of this help explain why otherwise repulsive electrons experience attraction, leading to superconductivity?
3. It is possible to stabilize the existence of Mott insulating ground states without the spontaneous breaking of any continuous symmetries, like in Neel antiferromagnets? If yes, is there anything in common between them and Quantum Spin Liquids on geometrically frustrated lattices?
4. Is it possible to build an overarching framework for the emergence of topologically ordered states of matter within systems of strongly correlated electrons that populate flat bands? If yes, would this shed light on the detection of fractionalised charge excitations in these systems?
5. Are there any universal patterns of many-particle entanglement in the various forms of quantum matter, and is it possible to probe these experimentally?
6. How does disorder interplay with electronic correlations in shaping quantum matter, and is knowledge of this going to be of use in the creation of novel quantum devices?

There is clearly a lot of opportunity in the field of quantum matter, both theoretical and experimental, and it will be exciting to see what the future holds.

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